SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1003	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	3-08-2026	**Duration:**	60 minutes

Code of Conduct

I __V.TEJASWINI(!92424360) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: __________

Annexure A – Scenario-Based Requirement Analysis

Smart **Waste Collection Monitoring and Route Optimization Platform**

Question

Based on the assigned problem statement, perform a Scenario-Based Requirement Analysis and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**

- Carefully understand the given problem statement.

Smart cities generate huge amounts of waste every day from residential areas, commercial buildings, industries, hospitals, and public places. Traditional waste collection methods follow fixed schedules without checking whether bins are full or empty. As a result, some bins overflow while others are collected unnecessarily. This leads to poor sanitation, unpleasant odors, increased fuel consumption, traffic congestion, and higher operational costs.

The Smart Waste Collection Monitoring and Route Optimization Platform is an IoT-based solution that monitors waste bin levels in real time and automatically plans the best collection routes. Smart sensors installed in waste bins measure the fill level and send data to a cloud server. The system analyzes this information and assigns optimized routes to waste collection vehicles. Authorities can monitor the entire waste management process through a web or mobile dashboard.

This system improves efficiency, reduces environmental pollution, saves fuel, minimizes collection time, and helps municipalities provide cleaner and healthier surroundings for citizens.

- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain, Stakeholders, Existing Challenges, and System Objectives

Business Domain

- The project belongs to the Smart City and Municipal Waste Management domain.

- It uses IoT sensors, GPS, Cloud Computing, AI, and Data Analytics to improve waste collection.
- The system monitors waste bin levels in real time and optimizes collection routes.
- It helps reduce fuel consumption, operational costs, traffic congestion, and environmental pollution.
- The platform supports efficient public services and sustainable urban development.

Stakeholders

- Municipal Corporation: Oversees waste management operations and monitors city cleanliness.
- Waste Collection Supervisors: Assign routes, manage vehicles, and monitor daily collection activities.
- Waste Collection Drivers: Follow optimized routes and update collection status after completing tasks.
- Sanitation Workers: Collect waste from smart bins and maintain cleanliness in public areas.
- Citizens: Report overflowing bins, track complaints, and provide feedback through the mobile application.
- System Administrator: Manages users, system configuration, security, and database maintenance.
- Government Authorities and Environmental Agencies: Monitor system performance and ensure compliance with environmental regulations.
- Maintenance Team: Repairs IoT sensors, GPS devices, and smart bins whenever technical issues occur.

Existing Challenges

- Waste collection follows fixed schedules instead of actual bin fill levels.
- Garbage bins often overflow, creating hygiene and environmental problems.
- Manual monitoring requires more time, effort, and workforce.

- Inefficient route planning increases fuel consumption and travel time.
- Lack of real-time monitoring makes it difficult to identify full bins quickly.
- Poor communication between citizens and municipal authorities delays issue resolution.
- Vehicle tracking and performance monitoring are limited in traditional systems.
- High operational costs and unnecessary collection trips reduce overall efficiency.

System Objectives

- Monitor waste bin levels continuously using IoT sensors.
- Automatically identify full bins and send alerts for collection.
- Generate optimized routes to reduce travel time, fuel usage, and operational costs.
- Track waste collection vehicles using GPS technology in real time.
- Improve waste collection efficiency and prevent garbage overflow.
- Provide dashboards, reports, and analytics for better decision-making.
- Allow citizens to report waste-related issues through a mobile application.
- Improve public sanitation, reduce environmental pollution, and support smart city initiatives.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

Functional requirements describe the core features and operations that the Smart Waste Collection Monitoring and Route Optimization Platform must perform.

- User Registration and Login: Allow administrators, municipal officers, supervisors, drivers, and citizens to securely register and log in to the system.
- User Role Management: Provide different access permissions based on user roles such as administrator, supervisor, driver, and citizen.
- Smart Bin Monitoring: Continuously monitor the waste level of smart bins using IoT sensors and display their current status.

- Bin Status Detection: Automatically detect bins that are full, nearly full, or empty and update the system in real time.
- Real-Time Notifications: Send alerts to supervisors and drivers whenever bins reach the predefined waste level.
- GPS Vehicle Tracking: Track the live location of waste collection vehicles and display their movement on a map.
- Route Optimization: Generate the shortest and most efficient collection routes based on waste levels, traffic conditions, and vehicle availability.
- Vehicle Assignment: Assign waste collection vehicles and drivers to optimized routes automatically or manually.
- Collection Status Update: Allow drivers to update the status of waste collection after servicing each bin.
- Citizen Complaint Management: Enable citizens to report overflowing bins, upload photos, and track complaint status.
- Dashboard and Analytics: Display real-time information about waste levels, vehicle locations, completed collections, and pending tasks.
- Report Generation: Generate daily, weekly, and monthly reports for waste collection, vehicle performance, and complaints.
- Maintenance Management: Monitor the health of IoT sensors and schedule maintenance for damaged smart bins or devices.
- Data Storage and Management: Store user details, sensor readings, collection records, complaints, and reports securely in the database.
- Notification Management: Send notifications through SMS, email, or mobile application regarding collection schedules and system alerts

o Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

Non-functional requirements define the quality attributes that ensure the system operates efficiently, securely, and reliably.

Performance

- The system should process sensor data and display updates in real time.
- Route optimization should be completed within a few seconds.
- The platform should support multiple users simultaneously without performance degradation.
- Reports and dashboards should load quickly with minimal delay.

Security

- All users must authenticate using secure login credentials.
- Sensitive data should be encrypted during storage and transmission.
- Role-based access control should prevent unauthorized access.
- Regular backups and audit logs should be maintained.

Reliability

- The system should provide accurate waste level monitoring.
- It should continue functioning even if individual sensors fail.
- Automatic data backup and recovery mechanisms should prevent data loss.
- The platform should maintain consistent performance with minimal downtime.

Usability

- The interface should be simple, user-friendly, and easy to navigate.
- The system should support both web and mobile applications.
- Dashboards should present information clearly using charts and maps.
- Users should be able to complete tasks with minimal training.

Scalability

- The system should support thousands of smart bins and multiple collection vehicles.
- It should be expandable to additional cities or municipalities.
- Cloud infrastructure should allow future storage and user growth.

- New features and modules should be integrated without major redesign.

Maintainability

- The software should have a modular architecture for easy updates.
- Bugs and system errors should be identified through logging mechanisms.
- System maintenance should be performed with minimal service interruption.
- Documentation should be available for developers and administrators.

Availability

- The platform should be available 24×7 for users.
- Cloud servers should provide high uptime and continuous monitoring.
- Disaster recovery mechanisms should restore services quickly after failures.
- The system should ensure uninterrupted access during peak operational hours.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

- System Administrator – Manages user accounts, configures the system, monitors overall performance, and generates reports.
- Municipal Officer – Monitors waste collection activities, views dashboards, and makes decisions based on reports.
- Waste Collection Supervisor – Assigns vehicles and drivers, creates collection schedules, and monitors daily operations.
- Waste Collection Driver – Views assigned routes, collects waste, and updates the collection status.
- Citizen – Reports overflowing bins, tracks complaint status, and provides feedback through the mobile application.

- Maintenance Technician – Repairs smart bins, replaces faulty sensors, and updates maintenance records.

Secondary Actors

- IoT Smart Bin Sensors – Continuously send waste level data to the system.
- GPS Tracking System – Provides the real-time location of collection vehicles.
- Cloud Database – Stores user information, sensor data, collection history, and reports.
- Notification Service (SMS/Email) – Sends alerts and notifications to users.
- Google Maps API – Supports route optimization and navigation for drivers.

  - List the major use cases associated with each actor.

System Administrator

- Register and manage user accounts for all stakeholders.
- Assign user roles and access permissions.
- Configure system settings and maintain the database.
- Monitor system performance and security.
- Generate administrative and operational reports.

Municipal Officer

- View the real-time dashboard of waste collection activities.
- Monitor the status of smart bins across the city.
- Track waste collection vehicles using GPS.
- Analyze reports and make operational decisions.
- Review citizen complaints and service performance.

Waste Collection Supervisor

- Assign optimized routes to collection vehicles.
- Allocate drivers based on schedules and availability.
- Monitor daily waste collection progress.

- Handle emergency collection requests.
- Ensure timely completion of collection tasks.

Waste Collection Driver

- Log in and view assigned collection routes.
- Navigate to waste bin locations using GPS.
- Collect waste from assigned smart bins.
- Update the collection status after completing each task.
- Report damaged bins or vehicle-related issues.

Citizen

- Register and log in to the mobile application.
- Report overflowing or damaged waste bins.
- Upload photos and location details with complaints.
- Track the status of submitted complaints.
- Provide feedback on waste collection services.

Maintenance Technician

- Receive notifications about faulty sensors or smart bins.
- Inspect and repair damaged smart bins.
- Replace defective IoT sensors and devices.
- Update maintenance records after completing repairs.
- Confirm that repaired bins are functioning properly.

o Draw a Use Case Diagram representing the interactions between actors and the system.

4. Prepare the Software Requirement Specification (SRS) Develop an SRS document containing:

- Introduction

The Smart Waste Collection Monitoring and Route Optimization Platform is a smart city solution designed to improve the efficiency of municipal waste collection using modern technologies such as IoT sensors, GPS tracking, Cloud Computing, and Data Analytics. The system continuously monitors the fill level of waste bins and automatically generates optimized routes for waste collection vehicles. It helps municipal authorities reduce operational costs, improve resource utilization, prevent garbage overflow, and maintain a clean and healthy environment. The platform also provides real-time monitoring, automated notifications, performance reports, and a citizen complaint management system

- Problem Description

The traditional waste collection system follows fixed schedules without considering the actual waste level in garbage bins. As a result, some bins overflow before collection while others are emptied even when they are not full. This causes unnecessary fuel consumption, increased operational costs, traffic congestion, poor sanitation, and environmental pollution. Manual monitoring also makes it difficult to track collection vehicles and respond quickly to citizen complaints. The proposed smart waste management system addresses these problems by providing real-time monitoring, intelligent route optimization, and automated waste collection management.

- Scope of the System

  - Monitor smart waste bins in real time using IoT sensors.
  - Detect full bins automatically and generate collection alerts.
  - Optimize collection routes to reduce travel distance and fuel consumption.
  - Track waste collection vehicles using GPS technology.
  - Allow citizens to report overflowing bins through a mobile application.
  - Provide dashboards for monitoring collection activities.
  - Generate daily, weekly, and monthly reports.
  - Maintain historical waste collection records.
  - mprove communication between citizens and municipal authorities.
  - Support smart city and sustainable waste management initiatives

- Functional Requirements

  - Monitor smart waste bins in real time using IoT sensors.
  - full bins automatically and generate collection alerts.
  - Optimize collection routes to reduce travel distance and fuel consumption.
  - Track waste collection vehicles using GPS technology. Allow citizens to report overflowing bins through a mobile application.

- Non-Functional Requirements

rformance

- Process real-time sensor data efficiently.
- Generate optimized routes quickly.
- Support multiple users simultaneously.

Security

- Secure user authentication.

- Data encryption during storage and transmission.
- Role-based access control.

Reliability

- Accurate sensor monitoring.
- Automatic data backup and recovery.
- Continuous system operation with minimal downtime.

Usability

- Simple and user-friendly interface.
- Easy navigation for all users.
- Responsive web and mobile applications.

Scalability

- Support thousands of smart bins and users.
- Expand to multiple cities without major changes.

Maintainability

- Easy software updates.
- Modular system architecture.
- Error logging and troubleshooting support.

Availability

- 24×7 system availability.
- Cloud-based deployment with high uptime.
- Disaster recovery support.

o Use Case Descriptions

o Actor: Citizen

- Purpose: Report an overflowing or damaged waste bin.
- Precondition: Citizen is registered and logged into the application.
- Main Flow:

Select the waste bin location.

Upload a photo (optional).

Submit the complaint.

System records the complaint and sends a notification.

- Postcondition: Municipal authorities receive the complaint for action.
- Use Case 2 – Generate Optimized Route
- Actor: Waste Collection Supervisor
- Purpose: Create the shortest and most efficient collection route.
- Precondition: Waste level data is available from smart bins.

Main Flow:

View bins requiring collection.

Generate optimized route.

Assign vehicle and driver.

Driver receives route details.

- Postcondition: Collection route is successfully assigned.
- Data Requirements

The system stores and manages the following information:

- User details (Administrator, Officers, Drivers, Citizens).
- Smart bin information and location.
- Waste level sensor readings.
- Vehicle and driver details.

- GPS tracking information.
- Collection schedules.
- Complaint records.
- Maintenance records.
- Notification history.

o External Interface Requirements

User Interface

- Web-based dashboard for administrators and officers.
- Mobile application for drivers and citizens.
- Responsive and easy-to-use interface.

Hardware Interface

- IoT smart waste bin sensors.
- GPS devices installed in collection vehicles.
- Smartphones and computers.

Software Interface

- Cloud database.
- Google Maps API for navigation.
- GPS tracking service.
- SMS and Email notification services.

Communication Interface

- Internet connection.
- Wi-Fi.
- 4G/5G mobile networks.
- HTTP/HTTPS communication protocols.

o System Features

- eal-time monitoring of smart waste bins.
- Automatic waste level detection using IoT sensors.
- Intelligent route optimization for waste collection vehicles.
- GPS-based vehicle tracking.
- Citizen complaint management system.
- Real-time notifications and alerts.
- Dashboard with charts and analytics.
- Vehicle and driver management.

5.Identify Assumptions and Constraints

o List the assumptions considered while developing the system.

- All waste bins are equipped with IoT sensors to monitor waste levels accurately.
- A stable internet connection is available for real-time communication between smart bins, vehicles, and the central server.
- GPS devices installed in waste collection vehicles provide accurate location information.
- Municipal authorities and waste collection staff are trained to use the system effectively.
- Citizens use the mobile application to report overflowing bins and track complaints.
- Cloud servers are available to securely store and process system data. Waste collection vehicles are available and maintained properly for daily operations.
- Sensor data received from smart bins is reliable and updated at regular intervals.
- Adequate power supply is available for IoT devices and communication systems.
- Government authorities support the implementation of smart waste management initiatives

o Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Technical Constraints

- Dependence on stable internet and network connectivity.

- Limited battery life and maintenance of IoT sensors.
- GPS signal accuracy may be affected in certain locations.
- Integration with external services such as Google Maps and cloud platforms.
- Limited storage and processing capacity if the system grows rapidly.

Operational Constraints

- Availability of trained municipal staff and vehicle drivers.
- Limited number of waste collection vehicles during peak operations.
- Regular maintenance required for sensors and smart bins.
- Delays caused by traffic congestion and unexpected road closures.
- Continuous monitoring is needed to ensure smooth system operation.

Economic Constraints

- High initial cost for installing smart bins and IoT sensors.
- Expenses for GPS devices, cloud services, and software development.
- Regular maintenance and replacement costs for hardware components.
- Budget limitations for system upgrades and expansion.
- Operational costs related to internet connectivity and technical support.

Legal Constraints

- Compliance with data privacy and cybersecurity regulations.
- Protection of personal information collected from users.
- Compliance with government environmental and waste management policies.
- Proper authorization for collecting and storing location data.
- Adherence to municipal rules and smart city standards.

Time-Related Constraints

- Limited project development and implementation schedule.

- Time required for system testing and quality assurance.
- Deadlines for deployment and user training.
- Periodic maintenance windows may temporarily affect system availability.
- Future enhancements must be completed within planned project timelines

6.**Requirement Completeness and Consistency**

- Verify whether all stakeholder requirements have been addressed.
  - Waste Collection Supervisors can assign routes, monitor vehicle movement, and manage daily collection schedules.
  - Waste Collection Drivers receive optimized routes, update collection status, and report issues during operations.
  - Citizens can register complaints, upload images of overflowing bins, track complaint status, and provide feedback.
  - System Administrators can manage users, configure system settings, monitor performance, and maintain security.
  - Maintenance Technicians can receive maintenance requests, repair faulty sensors, and update maintenance records.
  - Government Authorities can access performance reports and monitor environmental compliance.
  - Overall, the proposed system satisfies the functional needs of all major stakeholders involved in smart waste management.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.

Ambiguity

- Clearly define waste level thresholds (e.g., 80% or 90%) for generating collection alerts.
- Specify the response time for complaint handling and waste collection.
- Define user roles and permissions to avoid confusion.
- Clearly mention the frequency of sensor data updates.

Redundancy

- Remove duplicate notifications sent to multiple users.
- Avoid storing the same waste collection data in different modules.
- Prevent repeated complaint entries for the same waste bin.
- Eliminate duplicate report generation.

Inconsistency

- Maintain consistent terminology throughout the system documentation.
- Ensure that all modules use the same waste bin status values.
- Keep report formats and dashboard information consistent.
- Synchronize real-time data between sensors, vehicles, and the central database.

Missing Requirements

- Disaster recovery and backup mechanisms.
- Offline data synchronization during internet failures.
- Sensor calibration and health monitoring.
- Audit logs for user activities.
- Data retention and archival policy.
- Emergency notification and escalation procedures.
- API documentation for third-party integration.
- User training and help documentation

o Suggest improvements to enhance the quality and completeness of the requirement specification.

- Integrate Artificial Intelligence (AI) to predict waste generation and optimize collection schedules.
- Include real-time traffic information for better route optimization.

- Add predictive maintenance to identify faulty sensors before failure.
- Develop a multilingual mobile application to support more users.
- Provide voice-based complaint registration for easier accessibility.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate,	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram	Actors/use cases missing or incorrect; diagram absent or inaccurate.

	and follows UML standards.		partially correct.	
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25